Multi-Agent Mamba-RL □□□□□ □□□□□□□□□□ Pipeline □□□□

`MediaTek_2026_project`

2026 □ 8 □

Abstract

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1. □□□□□□□□□□ agent □□□□□□□□□ agent □□□□□□□coordinator □□□□□□ □□□□□□□
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2 □□□□□□□□

□□ pipeline □□ **??** □□□□□□□□□□□□□□□ launcher □□ import □□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□

Table 1: Multi-Agent Mamba-RL □□□□

□□	□□
`run_mamba_rl.sh`	□□□□□□GPU□cache □□□□□□□□
`src/data_mamba_rl.py`	MovieLens□Amazon□Yelp □ synthetic □□□□□□
`src/model_mamba_rl.py`	LoRA□selective-state specialist □ coordinator □□□
`src/train_mamba_rl.py`	□□□□□□REINFORCE□□□□□□□□□□□□□□□□
`tests/test_mamba_rl.py`	□□□□□agent □□□□□stage □□□□□ shape □□□
`outputs_mamba_rl/`	□ dataset/run ID □□□ checkpoint□metrics□□□□□□□ □□

3 □□□ Pipeline

□ **??** □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ full-catalog ranking □□□□□□ □□□□□□□□□□□□□□□

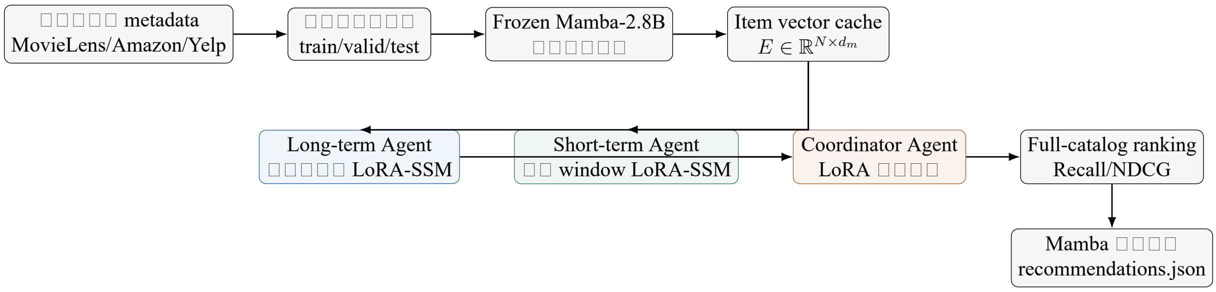

Figure 1: Multi-Agent Mamba-RL □□□□□□□

3.1 □□□□□□□□□□□□

□□□□□□ i□□□ title□genre□category□feature□description □ business metadata □□□□ t_i□ □□□□□ Mamba □□□

$$\boldsymbol{e}_i^{\text{text}} = \mathrm{MeanPool}(\mathrm{Mamba}_{\text{frozen}}(t_i)) \in \mathbb{R}^{d_m}. \tag{1}$$

□□□□□□□□□ $E_{\text{text}} \in \mathbb{R}^{N\times d_m}$ □□□ cache□□□□□ catalog □□□□□□□□ run □□□□□ cache□ □□□□□ 2.8B □□□

□□□□□□□□□□□

$$\boldsymbol{e}_i = \mathrm{norm}\left(W_p \boldsymbol{e}_i^{\text{text}}\right), \qquad W_p \in \mathbb{R}^{d\times d_m}, \tag{2}$$

□□□□□□□□□□□□□□ $d = 128$□W_p □□□ agent □□□ retrieval bridge□□ agent-specific adaptation □□□ LoRA □□□

4 □□□□□□□□□

4.1 □□□□□□□

□□ loader □□□□□□□□□

$$(u, i, \tau, t_i), \tag{3}$$

□□ u □□□□□ ID□i □□□ ID□τ □ timestamp□t_i □□□□□□ID □□□□□□□□□□□□□□□ □□□□□□□□□□□□□MovieLens □ Yelp □□□ rating/stars ≥ 4 □□□□ implicit feedback□

4.2 □□□□□□

Table 2: □□□□□□□□□□□□

□□□□□□	□□□□	□□□□
movielens-100k/1m/20m/25m/32m/latest-small	rating □ timestamp	□□ title □ genres□□□□□□□□□□ GroupLens □□□
amazon-<category>	Amazon Reviews 2023 review	metadata □ title□subtitle□main category□features□description□

□□□□□	□□□□	□□□□
amazon:<exact-config>	□□ Hugging Face raw review config	□□□ raw meta config □ parent_asin join□
yelp19, yelp23	Yelp review JSON □□□□□□	business name□categories□city□state □□□ attributes□
synthetic	□□□□□□□□□	□□□□□□□□□□ smoke test□

Yelp □□ snapshot □□□□□□□□□□□ launcher □□□□□□□□□ `DATA_PATH` □□□□□□□□CSV□JSONL □ Parquet□□□ Yelp JSON □□□ `yelp_academic_dataset_review.json` □ `yelp_academic_dataset_business.json`□

4.3 Chronological split

□□□□ u □□□□□□□□□

$$\mathcal{H}_u = (i_1, i_2, \ldots, i_{T_u}). \tag{4}$$

□□ leave-two-out□

$$\mathcal{H}_u^{\text{train}} = (i_1, \ldots, i_{T_u-2}), \tag{5}$$
$$i_u^{\text{valid}} = i_{T_u-1}, \tag{6}$$
$$i_u^{\text{test}} = i_{T_u}. \tag{7}$$

Validation □□□□ training history□test □□□□ training history □ validation item□□□□□□□□□□□□□□ state□

4.4 RL transition

training history □□□□ prefix-to-next-item transition□

$$((i_1, \ldots, i_k), i_{k+1}), \quad k = 1, \ldots, T_u - 3. \tag{8}$$

□ transition □□□□ `MAX_TRANSITIONS`□□□ reservoir sampling □□□□□□□□□ MovieLens-20M □□□ Amazon/Yelp □□□□□□□□□□□□ transition list□

5 Multi-Agent □□□□

5.1 Agent □□

□□□□□□□ agent □□□

Long-term Agent
□□□□ `max-history` □□□□□□□□□□□□□ time scale□0.08□□□□□□□□□□□□□□□□

Short-term Agent
□□□□□ `short-window` □□□□□□ 10□□□□□□ time scale□0.5□□□□□□□□ session intent□

Coordinator Agent
□□□□□□ state□□□□□□□□□□□□□□ mixing weights□□□□ specialist scores □□□□ coordinator score□

□□□□□□□ agent □□□□□ LoRA□□□□□ agent □□□□□□□□□□□□□□□□□□□□ GPU □□□□□□ Mamba-2.8B□□□□□□□□□□□□ agent specialization □□□□□□□□□□□□□□□□□□

5.2 LoRA □□□

□□□ $\boldsymbol{x} \in \mathbb{R}^{d_{in}}$□agent-specific LoRA update □□□□

$$\mathrm{LoRA}_j(\boldsymbol{x}) = \frac{\alpha}{r} B_j A_j \, \mathrm{Dropout}(\boldsymbol{x}), \tag{9}$$

□□ $A_j \in \mathbb{R}^{r \times d_{in}}$□$B_j \in \mathbb{R}^{d_{out} \times r}$□□□ $r = 8$□$\alpha = 16$□dropout $= 0.05$□A_j □□ Kaiming initialization□B_j □□□□□□□□□□□□□□ adapter □□□□□□ base path□

□□□□□□ agent □□□□ candidate□delta□gate□output □□ LoRA□coordinator □□ state□mix□score □□ LoRA□□□ agent □□□□□□□□□□□□□□□□ agent □ adapter□

5.3 Mamba □ selective state update

□□□ t □□□□□□□ $\boldsymbol{x}_t$□specialist $j \in \{L, S\}$ □□□

$$\boldsymbol{\delta}_t^{(j)} = \mathrm{softplus}\Big(\boldsymbol{b}_\delta^{(j)} + \mathrm{LoRA}_\delta^{(j)}(\boldsymbol{x}_t)\Big), \tag{10}$$

$$\boldsymbol{a}_t^{(j)} = \exp\Big(-\boldsymbol{\delta}_t^{(j)}\Big), \tag{11}$$

$$\tilde{\boldsymbol{h}}_t^{(j)} = \tanh\Big(\boldsymbol{x}_t + \mathrm{LoRA}_{\text{candidate}}^{(j)}(\boldsymbol{x}_t)\Big), \tag{12}$$

$$\boldsymbol{h}_t^{(j)} = \boldsymbol{a}_t^{(j)} \odot \boldsymbol{h}_{t-1}^{(j)} + (1 - \boldsymbol{a}_t^{(j)}) \odot \tilde{\boldsymbol{h}}_t^{(j)}, \tag{13}$$

$$\boldsymbol{g}_t^{(j)} = \mathrm{sigmoid}\Big(\mathrm{LoRA}_{\text{gate}}^{(j)}(\boldsymbol{x}_t)\Big), \tag{14}$$

$$\boldsymbol{o}_t^{(j)} = \boldsymbol{g}_t^{(j)} \odot \boldsymbol{h}_t^{(j)} + (1 - \boldsymbol{g}_t^{(j)}) \odot \boldsymbol{x}_t. \tag{15}$$

□□□□□ output LoRA □□ ℓ_2 normalization□□ recurrence □□□□□ T □□□□□□□ $\mathcal{O}(Td)$□□□□ Transformer □ $T \times T$ attention matrix□

$\boldsymbol{\delta}_t$ □ input-sensitive forgetting rate□□□□□□□□□□ state□□□□□□□□□□□□□□□ long/short agents □□□□□ time scale □□□□ observation window□□ specialization □□□□□ inductive bias□□□□□□ loss □□□□□□□

5.4 Specialist policy scores

□□□□□□□ $\boldsymbol{e}_i$□

$$z_i^{(j)} = \exp(\tau_j) \, \left\langle \boldsymbol{s}_u^{(j)}, \boldsymbol{e}_i \right\rangle, \tag{16}$$

□□ τ_j □□□□ log-temperature□□□□□ $\exp(\tau_j) \le 20$□normalized state □ item vector □□□□□ cosine similarity□

5.5 Coordinator □□□□

Coordinator □□□ specialist states □□□

$$\boldsymbol{q}_u = [\boldsymbol{s}_u^{(L)}; \boldsymbol{s}_u^{(S)}], \tag{17}$$

$$[w_L, w_S] = \mathrm{softmax}(\mathrm{LoRA}_{\mathrm{mix}}(\boldsymbol{q}_u)), \tag{18}$$

$$\boldsymbol{s}_u^{(C)} = \mathrm{norm}\big(w_L \boldsymbol{s}_u^{(L)} + w_S \boldsymbol{s}_u^{(S)} + \mathrm{LoRA}_{\mathrm{state}}(\boldsymbol{q}_u)\big). \tag{19}$$

□□□□□□□□

$$z_i^{\mathrm{final}} = z_i^{(C)} + w_L z_i^{(L)} + w_S z_i^{(S)}. \tag{20}$$

□□ coordinator □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

6 □□□□□□□

□□□□□□□□□□□□ reward □□□□ agent□□□□□ coordinator □□□□□□ specialist□□□□ specialist □□□□□□□□□□ pipeline □□□□□□□□

6.1 Stage 1□Specialist supervised warm-up

□□ long agent□short agent □□□ item projection□□□ coordinator□□□ transition □□□□□□□□□□□□□□ $K-1$ □□□□□□□ slate□

$$\mathcal{C} = \{i^+, i_1^-, \ldots, i_{K-1}^-\}, \qquad K = 64. \tag{21}$$

□□□□□□ index 0□loss □□

$$\mathcal{L}_{\mathrm{specialist}} = \mathrm{CE}(\boldsymbol{z}^{(L)}, 0) + \mathrm{CE}(\boldsymbol{z}^{(S)}, 0). \tag{22}$$

□□ agent □□□□ history window□□□□□ time scale□□□□□ target □□□□□□□□□□□□ state□

6.2 Stage 2□Coordinator warm-up

□□□□ specialist □ item projection□□□□ coordinator□

$$\mathcal{L}_{\mathrm{coord}} = \mathrm{CE}(\boldsymbol{z}^{\mathrm{final}}, 0). \tag{23}$$

□□ coordinator □□□□ specialist representation □□□□□□□□□ specialist □□□□□□□□□□

6.3 Stage 3□Joint multi-agent REINFORCE

□□□□ agent □□□ projection □□□□□□ policy □ sampled slate □□□□

$$\pi_j(a \mid s) = \mathrm{softmax}(\boldsymbol{z}^{(j)})_a, \qquad j \in \{L, S, C\}. \tag{24}$$

□ agent □□□ distribution □□ action□Reward □□□□

$$r_j = \begin{cases} 1, & a_j = 0, \\ -0.05, & a_j \neq 0. \end{cases} \tag{25}$$

□□ agent □□□□□ exponential moving-average baseline□

$$b_j \leftarrow 0.95 b_j + 0.05 \bar{r}_j, \qquad A_j = r_j - b_j. \tag{26}$$

Policy-gradient □□□

$$\mathcal{L}_{\text{PG}} = -\sum_{j \in \{L,S,C\}} \mathbb{E}\left[A_j \log \pi_j(a_j \mid s_j)\right] - \beta \sum_j \mathbb{E}\left[H(\pi_j)\right], \tag{27}$$

□□ entropy coefficient $\beta = 0.01$□Advantage □□□□□□ detach□gradient □□ `log_prob` □ □ logits □ LoRA□□□□□□□ `sample()`□

□□□ sampled-policy training□joint stage □□□□ supervised anchor□

$$\mathcal{L}_{\text{sup}} = \frac{1}{3} \sum_{j \in \{L,S,C\}} \text{CE}(\boldsymbol{z}^{(j)}, 0). \tag{28}$$

□□□ long/short collapse□□□ specialization regularizer□

$$\mathcal{L}_{\text{spec}} = \cos^2\left(\boldsymbol{s}_u^{(L)}, \boldsymbol{s}_u^{(S)}\right). \tag{29}$$

□□ joint loss□

$$\boxed{\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{PG}} + \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}}} \tag{30}$$

□□ $\lambda_{\text{sup}} = 0.1$□$\lambda_{\text{spec}} = 0.01$□

6.4 Optimizer□AMP □□□□□□

□□ stage □□□□ AdamW optimizer□□□ learning rates □□

Stage	Learning rate	□□□□□
Specialists	2×10^{-4}	long□short□item projection
Coordinator	2×10^{-4}	coordinator
Joint RL	5×10^{-5}	□ agent □ item projection

CUDA □□□□ FP16 autocast □ GradScaler□□□ optimizer step □□ global gradient norm □ □□ 1.0□□□□ joint learning rate □□□ RL □□□□ retrieval space □□□□

7 Gradient □ Credit Assignment

Coordinator loss □□□□□□□□ long/short states □ scores□Joint stage □□□ gradient □□□ □□□□ states □□□ specialist LoRA□

$$\frac{\partial \mathcal{L}_C}{\partial \theta_L} = \frac{\partial \mathcal{L}_C}{\partial z^{\text{final}}} \frac{\partial z^{\text{final}}}{\partial \boldsymbol{s}^{(L)}} \frac{\partial \boldsymbol{s}^{(L)}}{\partial \theta_L}, \tag{31}$$

short agent □□□□□□□□ specialist □□□□□ sampled action□reward □ policy loss□□□□ □□□□ coordinator □□□□□□

□□ credit assignment □□ agent-local next-item reward □□□□□□□□□□□□□□□□□□ □□□□□□□□□□□ COMA/QMIX □ counterfactual critic□□□□□□□□ simulator□□□□□□ □□

$$A_j^{\text{cf}} = Q(s, \boldsymbol{a}) - Q(s, \boldsymbol{a}_{-j}, a_j^{\text{baseline}}), \tag{32}$$

□□□□□□□ agent □□□ reward □□□□□□

8 □□□□□□□□□□□

8.1 Full-catalog scoring

Evaluation □□□ 64-item sampled slate□□□□□□□□□□□□□□□

$$E = W_p E_{\text{text}} \in \mathbb{R}^{N \times d}, \tag{33}$$

□□□□□□□□□□□□□□□□□□□□□□

$$\boldsymbol{z}_u = E \boldsymbol{s}_u. \tag{34}$$

□□□□□ $B \times N \times d$ □□□□□□□□ $B \times d$ □ $d \times N$ □ GEMM□□□ catalog ranking □□□□□□□ GPU □□□□□

8.2 Seen-item masking

□ evaluation history □□□□□□ target □□□□□

$$z_{u,i} = -\infty, \tag{35}$$

□□□□□□□□□□□□□Validation history □ train□test history □ train □ valid□

8.3 Metrics

□□ target rank □□□□

$$\text{rank}_u = \sum_{i=1}^{N} \mathbb{I}[z_{u,i} \geq z_{u,i^+}]. \tag{36}$$

□□□□□

$$\text{Recall@K} = \frac{1}{|U|} \sum_u \mathbb{I}[\text{rank}_u \leq K], \tag{37}$$

$$\text{NDCG@K} = \frac{1}{|U|} \sum_u \frac{\mathbb{I}[\text{rank}_u \leq K]}{\log_2(\text{rank}_u + 1)}, \tag{38}$$

□□ $K \in \{5, 10\}$□□□□□ users/s □ user–item scores/s□□□□□□□□□□□□□□ catalog scoring □□□□

9 □□□□□□□□□

1. □□□□□□□□□□Mamba-2.8B □□□ dataset□item count □□□ fingerprint □□□cache hit □□□□□□□□□□
2. □□ **base semantics**□□ agent □□□□□□□ projection□□□□□□ 2.8B weights □□□ catalog embeddings□
3. □□ **history update**□specialist recurrence □□□□□□□□□□□□memory □□□ T^2 attention map□

4. □□ **history**□□□□□ 100 □□short agent □□ 10 □□□□□ latency□
5. □□□□□□□□□□□□□□□ 64-item slate □□ policy□evaluation □□ GEMM □□□ catalog scoring□
6. **Mixed precision**□Mamba cache □ CUDA □□□□ FP16□GradScaler □□□ gradient underflow□
7. **Reservoir transition cap**□□□□□ 500,000 transitions□□□□□□□□□ Python object memory □□□□□□
8. □□□□ **reason generator**□□□□□□□□□ checkpoint □□□□□□ policy □□ CPU□□□ CUDA cache□□□□□□□□□

10 □□□□□□

□□□□□□□□□□□□□□□ latency □□ full-catalog scoring□□□□ test users□□□□ Top-1 item□□□□□□□□□□□□ coordinator □□□ w_L, w_S□□□□□

Listing 1: □□□□ prompt □□□□□

```
Explain this recommendation in one concise sentence using only the
   supplied history.
Long-term agent weight=<wL>; short-term agent weight=<wS>.
History: <recent item texts>.
Recommended item: <recommended item text>. Reason:
```

Mamba □ greedy decoding□`do_sample=False`□□□□□ 40 □□ token□□□□□□□ agent weights □□□□ `recommendations.json`□□□ post-hoc grounded explanation□□□ reason quality □□□□ RL reward□□□□□□□□□□□□□□□□ reasoning-aware recommendation□□□□ item consistency□history grounding □□□□□□ reward□

11 □□□□□□□

11.1 Launcher

```
# MovieLens
bash run_mamba_rl.sh movielens-1m 0
bash run_mamba_rl.sh movielens-20m 0

# Amazon Reviews 2023
bash run_mamba_rl.sh amazon-all-beauty 0
bash run_mamba_rl.sh amazon-movies-and-tv 0

# Yelp licensed local snapshots
bash run_mamba_rl.sh yelp19 0 /data/yelp-2019
bash run_mamba_rl.sh yelp23 0 /data/yelp-2023
```

Launcher □□□□□□□□□□ `DATASET`□`CUDA_ID`□`DATA_PATH`□MovieLens □□□□□□Amazon □ Hugging Face cache □□□Yelp □□□□□□□

11.2 □□□□□□

Table 3: `run_mamba_rl.sh` □□□□□

□□	□□□	□□
CACHE_DIR	/workspace/P78123011/cache	HF□□□□ Mamba vector cache□
BATCH_SIZE	128	□□□□□ batch size□
EVAL_BATCH_SIZE	128	Full-catalog evaluation batch□
SPECIALIST_EPOCHS	1	Stage 1 epoch □□
COORDINATOR_EPOCHS	1	Stage 2 epoch □□
RL_EPOCHS	3	Joint REINFORCE epoch □□
MAX_TRANSITIONS	500000	Prefix transitions □□□
SEED	25252	Python□NumPy □ PyTorch seed□

□□□

```
BATCH_SIZE=256 RL_EPOCHS=5 MAX_TRANSITIONS=1000000 \
  bash run_mamba_rl.sh movielens-20m 0
```

11.3 □□□ CPU smoke test

```
python -m src.train_mamba_rl \
  --dataset synthetic \
  --device cpu \
  --skip-mamba \
  --no-generate-reasons \
  --max-transitions 32 \
  --batch-size 8
```

`--skip-mamba` □□□□ seed □ random vectors □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

12 □□□□□□□□

□□ run □□□

```
outputs_mamba_rl/<dataset>/<YYYYMMDD_HHMMSS>/
|-- train.log
|-- multi_agent_lora.pt
|-- loss.png
|-- metrics.json
`-- recommendations.json
```

`multi_agent_lora.pt` □□□□ state dict□□□ CLI config□validation/test metrics □ item-vector artifact path□□□ item feature buffer □□□□□ checkpoint□□□□□□□ artifact path □□□□□ cached item matrix□□□□ state dict□

`recommendations.json` □□ user index□□□ item indices□held-out target□Top-10□long/short agent weights□Top-1 □□□□□□□□□□`metrics.json` □□□□ validation □ test □□□□□□□□□□

13 □□□□□□□□□□□□□□

1. □□ **next-item reward**□□□ held-out item □ +1 reward □□□□□□□□□□□□□□□□□ □ -0.05 □□□□□□□□□□□□□□□□
2. □□□□□□□ **MDP**□Amazon/MovieLens/Yelp □□ log □□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□ cooperative contextual-bandit/ sequential-ranking formulation□
3. □□□□□□□□□□□□□□□□□□ logging policy □□□□□□□□□□□□□□□□□□ propensity score □ off-policy correction□
4. **Local reward credit**□□□ specialist □□□ action □□ reward□□□□□ centralized critic □ counterfactual advantage□
5. □□–□□□□□□□□□□ 64-item slate□□□□□□□□ catalog□□□□ full-catalog negatives □□□□□□□□□□□
6. □□□ **post-hoc**□□□□□□□□□□□□□□□□□□□□□□□□□□
7. **Mamba** □□□□□□□□ Mamba □□□□ semantics□□□□□□□ Mamba-inspired diagonal selective SSM□□□□□□□□ 2.8B Mamba adapters□

□□ Recall/NDCG □□□

14 □□□□□□□□□

□□□ multi-agent □ Mamba-specific □□□□□□□□□□□□□ **??** □□ ablation□

Table 4: □□ ablation □□□□□□

□□	□□□□
□ agent vs. long+short+coordinator	Multi-agent □□□□□□□□□□□□□□□□□
□□ window vs. 100/10 windows	□□□□□□□□□□□□□□□□
□□ 0.5/0.5 vs. learned coordinator	□□□□□□□□□□□□□□□
□ Stage 1/2□□□ joint RL	□□□□□□□□□□ variance □ collapse□
□ specialization loss	LoRA agents □□□□□□□□□□ states□
□ supervised anchor	Pure REINFORCE □□□□□□□□
Mamba text vector vs. random/ID embedding	□□□□□□ cold/sparse item □□□□
Selective recurrence vs. GRU/-Transformer	□□□history scaling□latency □ memory □□□□
Sampled vs. hard negatives	Full-catalog ranking gap □□□□□
□□□ reason generation	□□ throughput □□□□□□□□□□□□□□

□ Recall/NDCG □□□□□□□ catalog coverage□novelty□intra-list diversity□□□ history □□□□□□□□□agent weight □□□state cosine similarity□training transitions/s□evaluation users/s□scores/s□peak GPU memory □ reason latency□

15 □□□□□□□□

15.1 Reward-adaptive Mamba dynamics

□□ $\boldsymbol{\delta}_t$ □□□□□□□□□□□□□□ feedback□dwell time □ uncertainty □□□

$$\boldsymbol{\delta}_t = f_\theta(\boldsymbol{x}_t, r_{t-1}, \sigma_{t-1}), \tag{39}$$

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ reward-conditioned state dynamics□

15.2 Offline RL □□□□□

□ logged recommendation □□ behavior policy propensity□□□□ Conservative Q-Learning□

$$\mathcal{L}_{\mathrm{CQL}} = \log \sum_a \exp Q(s, a) - Q(s, a_{\mathrm{logged}}), \tag{40}$$

□□□ out-of-distribution □□□□□□□□□

15.3 Centralized critic □□□ simulator

□ KuaiSim□RL4RS □□□□□□□□□□□□ centralized training/decentralized execution□□ critic □□□□□ agent actions□□□ session watch time□retention□diversity □□□ reward □□□□□ hit reward□

15.4 Reasoning-aware recommendation

□□ policy □□□□ item □ reasoning trajectory□□□□

$$r = \lambda_1 r_{\text{rank}} + \lambda_2 r_{\text{grounding}} + \lambda_3 r_{\text{consistency}} + \lambda_4 r_{\text{diversity}}, \tag{41}$$

□□ group-relative policy optimization □□□□□□ trajectory □ gradient variance□

16 □□

Multi-Agent Mamba-RL pipeline □□□ Mamba □□□□□□Mamba □□□ selective state□ agent-specific LoRA □ policy gradient □□□□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□ selective-state agents □□□□□□□□□□□□ □ coordinator □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ RL □ □□□□□□□ GEMM □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ multi-agent recommendation □□□□□□□□□□□□□□□ reward-adaptive dynamics□offline RL□ centralized critic □ reasoning reward □□□□